



FRANCK MESSOU

<https://mesabo.github.io>



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Tokyo Koganei, Japan

LANGUAGES

Japanese: Simple conversations
English: Fluent
French: Native

SKILLS

Abilities

- ✦ Rigorous Crative
- ✦ Analytic Respectful
- ✦ Adaptable Open mind

Skills

- ✦ Programming Languages
 - Java, Python, Dart, Bash, C
- ✦ Artificial Intelligence
 - Deep learning, LLM, IoT, Cloud GPU
- ✦ Databases
 - SQL, NoSQL (MongoDB, Oracle)
- ✦ Architecture & Dev
 - APIs, Prototyping, Designing

INTERESTS

- 🎵 Music
- ⚽ Football
- 💃 Dance

Passionate about Data Science, applied AI, systems design, and software engineering, I specialize in **Python, Java, Flutter, SQL**, advanced **algorithms**, big data analysis, and scalable **backend** architectures. Leveraging strong analytical skills, engineering fundamentals, and technical precision, I create efficient, scalable, **production-ready AI** and software solutions — from cross-platform **mobile apps** and **REST APIs** to IoT-integrated systems, **big-data pipelines**, and data-driven platforms — tailored to solve complex challenges in IoT sensors, Big Data, and modern applications.

WORK EXPERIENCES

FullStack Developer & AI

[D'isum Inc.](#) (Part-time Tokyo)

Sep 2025 - Jun 2026

- **Sensor Data Pipeline Architect & Analyst** (Emphasizes building pipelines from raw sensor data);
- **Sensor Data Specialist:** App Development, Prototyping & Integration (Covers full scope: development, architecture/prototypes, team collaboration on core software integration.)
- **Precision Sensor Data Engineer - Analytics, ML Prototypes & System Fusion** (Incorporates ML/predictive elements, prototyping, and integrating new systems with existing ones.)

AI Research and Development

Jul 2024

Internship - [Japan AI Consulting](#), Osaka

- Worked with Generative AI, integrating Large Language Models (**LLM**) with backend API (**Flask**) and Database (**log files + PostgreSQL**) using Cloud Dev.

FullStack Developer

[Casys Technologies](#) (Full-time)

Jan 2023 - Aug 2023

- Improvement of backend systems in Java for commercial utility applications. (**CI/CD — Agile**)
- Integrated backend systems with IoT devices such as vehicle sensors and door captors, enhancing system functionality.

BackEnd & FronEnd Developer

ANOHOU Solution (Part-time, Ivory Coast)

Au 2019 — Sep 2022

- Developed systems , improving system performance and optimizing data management (**Flutter, FastApi, NodeJs, GraphQL, MySQL, MongoDB**).
- Member to the development of [EKKOU](#) app with ANOHOU Solutions, employing agile methodologies to deliver scalable solutions (Flutter)

PUBLICATIONS

Research Assistance at YU Lab - Tokyo (2025 – 2026)

Advancing Optimization through Lightweight LLM for IoT-enabled smart grids.

Research Machine Learning and Data Analysis (2024 – 2025)

International Collaboration for Enhancing Transformer-based models for improved performance in Energy Management solutions.

EDUCATION

Ph.D DC1 in Applied Informatics

Early graduation – April 2027

Hosei University – Tokyo, Japan

Master 2 in Applied Informatics

Sep 2023 – Sep 2025

Hosei University – Tokyo, Japan

PAPERS

- ***TSFormer: Temporal-Aware Transformer for Multi-Horizon Forecasting with Learnable Positional Encodings and Attention Mechanisms***, The 16th ICUFN 2025
- ***Federated Fine-Tuning of Large Language Models for Intelligent Automotive Systems with Low-Rank Adaptation***, IEEE VTC2025-Spring.
- ***Dynamic IoT Resource Allocation Using Graph Reinforcement Learning***, IEEE VTC2025-Spring.
- ***Enhancing Federated Learning in Consumer Electronics with Decoupled Knowledge Distillation against Data Poisoning***, 2025 IEEE 22nd CCNC, doi: 10.1109/CCNC54725.2025.10976151.
- ***Enhancing Short-Term Load Forecasting in Internet of Things: A Hybrid Attention-based CNN-BiLSTM with Data Augmentation Approach***, IEEE VTC2024-Fall.
- ***A Byzantine-Fault-Tolerant Federated Learning Method Using Tree-Decentralized Network and Knowledge Distillation for Internet of Vehicles***, IEEE VTC2024-Fall.
- ***A Resource-efficient Text-to-Text Transfer Transformer Encoder-based Vertical Hybrid Model for Malicious URLs Detection***, IEEE VTC2024-Fall.